



NATIONAL UNIVERSITY OF MODERN LANGUAGES

SOFTWARE QUALITY ENGINEERING LAB REPORT – 2

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PROGRAM: BSSE (6th Semester)

COURSE: SOFTWARE QUALITY ENGINEERING

SUBMITTED TO: Mam Rubab

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DAY: Saturday

Lab Report: Program Analysis and Defect Classification

Given Program

```
#include <iostream>
using namespace std;

int main() {

    int marks;

    cout << "Enter student marks (0-100): ";
    cin << marks;

    if (marks == 90)
        cout << "Grade: A";

    else if (marks > 80)
        cout << "Grade: B";

    else if (marks > 70)
        cout << "Grade: C";

    else if (marks > 60)
        cout << "Grade: D";

    else
        cout << "Grade: F";

    return 0;
}
```

Task #1 — Compile and Observe

1. Wrong Input Operator (Line 07)

Problem: cin uses >> operator for input, not <<.

Incorrect: cin << marks;

Correct: cin >> marks;

2. Wrong Relational Operator (Line 09)

Problem: == is invalid in C++.

Incorrect: marks == 90

Correct: marks >= 90

Corrected Program

```
#include <iostream>
using namespace std;

int main() {

    int marks;

    cout << "Enter student marks (0-100): ";
    cin >> marks;

    if (marks >= 90)
        cout << "Grade: A";

    else if (marks > 80)
        cout << "Grade: B";

    else if (marks > 70)
        cout << "Grade: C";

    else if (marks > 60)
        cout << "Grade: D";

    else
        cout << "Grade: F";

    return 0;
}
```

Task #2 — Identify and Classify Issues

Error: Human mistake made during coding.

Fault: Incorrect statement present in the program.

Failure: Incorrect behavior observed during execution or compilation.

Defect No	Line No(s)	Observed Failure	Fault in Code	Human Error (Mistake)	Category

1	07	Program fails to compile due to invalid input statement	cin << marks;	Used output operator instead of input operator	Fault
2	07	Compilation error related to operator overloading	Wrong operator with cin	Confusion between cin and cout operators	Failure
3	09	Program fails to compile due to invalid relational operator	marks ==> 90	Typed incorrect comparison operator	Fault
4	09	Compiler shows syntax error near >	Invalid syntax ==>	Lack of attention to syntax	Failure
5	07,09	Entire grading system cannot execute	Syntax errors stop compilation	Coding mistakes during implementation	Error

Conclusion

The program contained syntax faults, human errors, and compilation failures. The main problems were incorrect stream operators and invalid relational operators. After correcting them, the program executed successfully.